

4(a)	270°	A1
4(b)	arrow pointing vertically downwards at T	A1
4(c)	$v = f\lambda$ or $v = \lambda / T$ and $f = 1 / T$	C1
	wavelength = $0.62 \times 10^{-2} \times (4 / 3)$ (= $0.83 \times 10^{-2} \text{ m}$)	C1
	$f = 0.27 / (0.83 \times 10^{-2})$ = 33 Hz	A1
4(d)	resultant displacement is the sum of the displacements of the waves (from S_1 and S_2) or waves (from S_1 and S_2) <u>superpose</u> (at P)	B1
	<i>Any one point from:</i> (at P) the waves (from the two sources) (always) destructively interfere (at P) the waves have a path difference that is (always) an odd number of half-wavelengths / their path difference is one and a half wavelengths (at P) the waves have a phase difference that is (always) 180° / they are in antiphase / crest of one wave meets trough of other wave	B1
	amplitude (of the resultant wave) is zero (at all times)	B1